WEEK 1 · LESSON 3 OF 24 · HANDOUT

Intelligent Agents and Environments

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=BkedAnQfJ_U — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50 AI, Lecture 1: Knowledge — https://www.youtube.com/watch?v=HWQLez87vqM

Your goal in this lesson: Describe any AI system as an agent, use the PEAS method to specify it, and classify its environment.

What is an agent?

An **agent** is anything that **perceives** its environment through **sensors** and **acts** on it through **actuators**.

- A person: eyes and ears are sensors; hands and voice are actuators.
- A self-driving car: cameras and radar are sensors; steering and brakes are actuators.
- A chatbot: your typed message is its input (sensor); its reply is its action.

An agent's **percept** is what it senses at one moment. Its **agent function** maps what it has perceived so far to an action. The goal of AI is to build **rational agents**: agents that choose the action expected to give the best result, given what they know.

Rational does not mean perfect. A rational agent can still be wrong when it lacks information — just like a careful doctor who has not yet seen a test result.

Specifying an agent with PEAS

Before building an agent, engineers describe its task with **PEAS**:

Letter	Question	Example: a delivery drone in Rwanda
Performance measure	How do we judge success?	On-time delivery, safety, battery use, cost
Environment	Where does it work?	Hills, wind, rain, airspace, hospitals
Actuators	How does it act?	Motors, parachute release, radio messages
Sensors	How does it perceive?	GPS, cameras, wind and battery sensors

Getting the **performance measure** right is the most important step. If you reward a drone only for speed, it may take unsafe risks. What you measure is what you get.

Kinds of environments

Property	One side	Other side
Observable	**Fully** — the agent sees everything it needs (chess board)	**Partially** — some things are hidden (a driver cannot see round a corner)
Outcome	**Deterministic** — the same action always gives the same result	**Stochastic** — chance plays a part (weather, other people)
Time	**Episodic** — each decision stands alone (sorting photos)	**Sequential** — today's action changes tomorrow (driving, investing)
Change	**Static** — the world waits while the agent thinks	**Dynamic** — the world keeps moving (traffic)
Agents	**Single-agent**	**Multi-agent** — others act too (football, markets)

The harder side of each row makes the agent harder to build. A self-driving car in Kigali traffic faces almost all the hard sides: partially observable, stochastic, sequential, dynamic and multi-agent.

Types of agents (simplest to most capable)

1. **Simple reflex agent** — reacts to the current percept with a rule (*if the room is dark, turn on the light*).
2. **Model-based agent** — keeps an internal picture of the world, so it can handle what it cannot see right now.
3. **Goal-based agent** — plans actions to reach a goal (*reach the hospital*).
4. **Utility-based agent** — compares how *good* different outcomes are (*reach the hospital fast, safely and cheaply*).
5. **Learning agent** — improves its own behaviour from experience.

Remember this: Any AI system can be described as an agent that senses and acts. PEAS makes the task clear, and the type of environment tells you how hard the problem is.

Name: _____ Date: _____

Activity

Task 1

Choose one system: a MoMo fraud detector, a crop-disease app, a customer-service chatbot or a traffic-light controller.

Task 2

Write its full PEAS table.

Task 3

Classify its environment on all five properties and explain each choice in one sentence.

Task 4

Say which type of agent it most likely is.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What is an agent in AI?

   A. A database
   B. Only a human customer-service worker
   C. Anything that perceives its environment through sensors and acts through actuators
   D. A type of computer virus

2. In PEAS, what does the P stand for?

   A. Performance measure
   B. Power
   C. Prediction
   D. Program

3. A driver cannot see what is round the next corner. This environment is…

   A. Single-agent
   B. Partially observable
   C. Fully observable
   D. Static

4. Which agent compares how good different outcomes are, not just whether a goal is reached?

   A. Episodic agent
   B. Utility-based agent
   C. Model-based agent
   D. Simple reflex agent

5. What does 'rational agent' mean?

   A. An agent that chooses the action expected to give the best result, given what it knows
   B. An agent with feelings
   C. An agent with no sensors
   D. An agent that is never wrong

6. Why is the performance measure so important?

   A. It decides the colour of the robot
   B. It replaces the sensors
   C. The agent will pursue whatever is measured, so a bad measure leads to bad behaviour
   D. It is only needed for games

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